

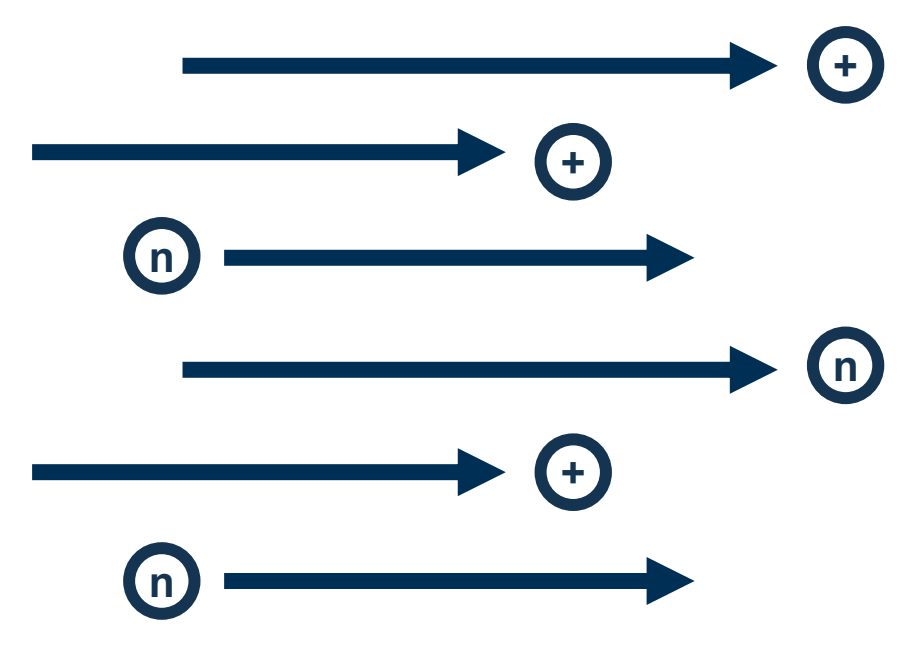
Remote, non-destructive quantification of SiC damage and lifetimes using Raman spectroscopy

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Silicon Carbide (SiC) is a leading candidate material for fusion blanket and nuclear structural components due to high thermal conductivity, low neutron activation and radiation resistance.



Irradiation damage (defects, amorphization, carbon clustering...) **degrades performance and limits component lifetimes.**

Raman spectroscopy provides a **non-destructive probe of lattice vibrations**, tracking phonon peak broadening, shifting, and carbon band emergence.

The project fostered our research skills and led to a final report with data and analysis with potential for publication and broader impact.

We looked to:

- **Quantify** irradiation damage
- **Relate** Raman signatures to disorder and amorphization
- **Assess** potential for non-destructive lifetime monitoring

Conditions of Analysis

Material: 6H-SiC single crystals

Ion species: Ne, Si and Au ions

Dose levels: Low damage (0.25 dpa) and high damage (2.5 dpa)

Temperatures: 300 & 750 °C

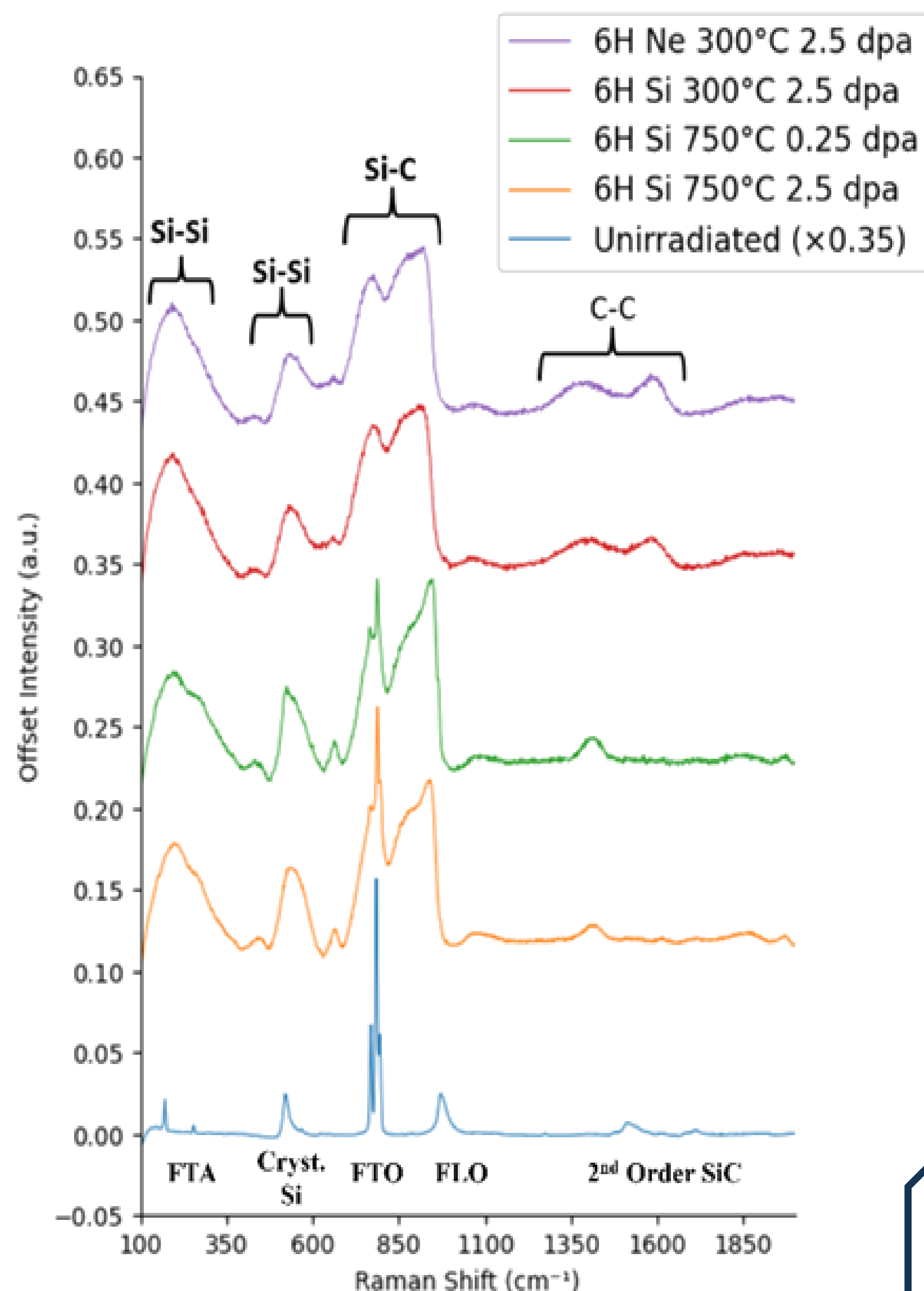
Annealing: In-situ in 100 °C steps up to 1200 °C

Purpose was to simulate neutron damage regimes, compare light vs heavy ion effects, chemical effects and recrystallisation

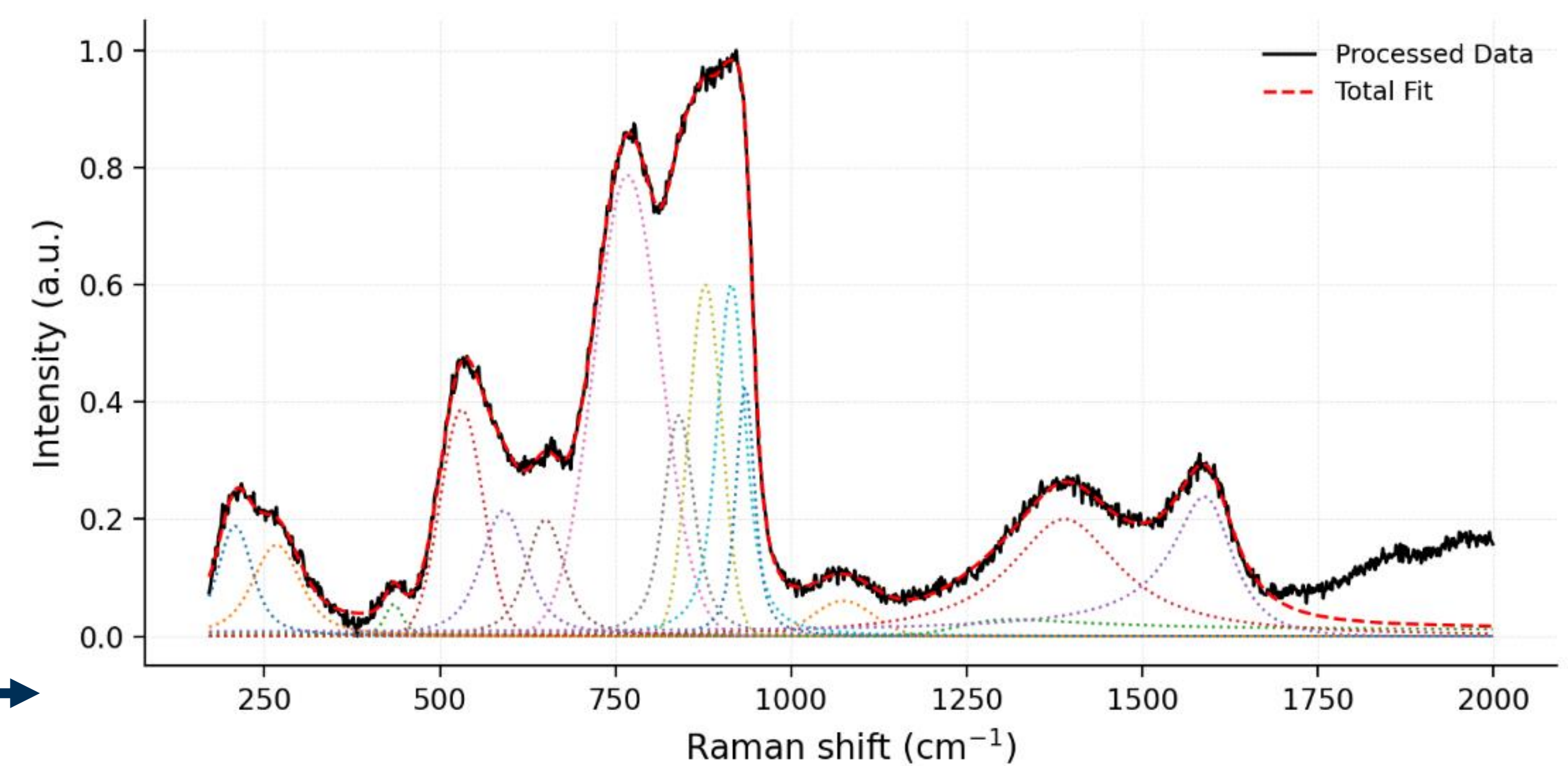
Through a *Renishaw inVia* confocal Raman microscope with 5x, 20x and 100x magnification, we observed reaction-bonded (RB) SiC: first in a pristine state (as shown) then irradiated at different fluences and temperatures with varying ion species.

In the 100x image shown of RB-SiC, regions marked 1, 2 & 3 mark pores, SiC "islands" and crystalline Si regions

Our Raman setup consisted of a 532 nm laser, 100x objective, and 0.3 cm⁻¹ resolution over ~100-2000 cm⁻¹, calibrated to the Si peak at 520 cm⁻¹



Curve fitting the resulting spectra involved processing raw data and using approximations with mathematical models with our Python scripts. Each peak, set with physical grounding, corresponds to a particular bond's vibration which we can identify (Si-C, Si-Si, C-C...)



Overlay of spectra of pristine (bottom) and irradiated SiC highlight the spectral signatures which correspond to physical changes. Broadening of the crystalline peaks is evident, with new peaks due to homonuclear (Si-Si & C-C) bonds emerging with damage.

Through literature review and experimental data, we explored trends relating temperature, fluence and chemical effects to disorder, amorphization and recrystallization. For example, we tracked chemical disorder as a function of the annealing temperature

